TEAM 9 • PES UNIVERSITY

SOFTWARE REQUIREMENTS SPECIFICATION

Electronics Store Inventory Management System

Updated requirements baseline and acceptance criteria

Document control	Details
Document ID	EIMS-SRS-1.1
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Revision purpose. Preserve the September 2026 baseline while extending the requirements, design and verification coverage to include functionality added during subsequent development sprints.

Revision history

Version	Date	Change summary	Approval
1.0	06-09-2026	Original baseline: inventory, reports, dashboard, authentication and security.	Pending in source document
1.1	03-11-2026	Adds Sprint 2 and subsequent enhancements: suppliers, branches, stock movements, purchase orders, image lifecycle, monitoring, UI and API error handling. Original IDs retained.	For team/guide review

Approvals

Role	Name	Signature / date
Project Team Lead / Product Owner	Karan Maheshwari	To be completed
Development Lead	Krishitha Kankanala	To be completed
QA Lead	Hemavathi E	To be completed
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1. Introduction

1.1 Purpose. This SRS defines the functional, non-functional, interface and security requirements for EIMS. Version 1.1 retains the original September baseline and adds requirements arising from the implemented project enhancements.

1.2 Scope. EIMS is a web-based electronics-store inventory application. The current implementation includes component inventory, supplier and branch administration, stock movement history, purchase order workflows, availability reporting, dashboard functions, component image management, authentication/RBAC, and system availability reporting. The project does not claim point-of-sale billing or external payment-gateway integration.

1.3 Audience. Development team, QA/test team, project guide/evaluator, store administrators/managers and store staff.

1.4 Definitions. EIMS—Electronics Store Inventory Management System; SKU—Stock Keeping Unit; RBAC—Role-Based Access Control; REST—Representational State Transfer; API—Application Programming Interface; PO—Purchase Order; RTM—Requirements Traceability Matrix; STP—Software Test Plan.

1.5 Document conventions. “Shall” denotes a required system behaviour. Requirement IDs are stable identifiers. Acceptance criteria are observable checks; test IDs are defined in the companion Test Plan.

2. Overall description

2.1 Product perspective. The implemented product is a layered web application with a browser-based frontend, a FastAPI REST backend, SQLAlchemy persistence and a SQLite database in the documented local configuration. The architecture separates presentation, API/application logic and data access.

2.2 Product functions. Core functions include inventory listing/search/filter/sort/update; component creation and deletion; supplier and branch management; image upload and lifecycle management; stock-in/stock-out history; purchase order creation and receiving; availability reports and exports; dashboard summaries; authentication and role-based access; and system availability reporting.

2.3 User classes. Admin/Store Manager users manage inventory and protected workflows. Staff users access permitted inventory and read operations; specific write permissions are enforced by backend authorization. The application must enforce authorization at the API, not only hide frontend controls.

2.4 Operating environment. Modern desktop and tablet browsers; application server running the Python/FastAPI stack; relational database via SQLAlchemy. Actual deployment security (HTTPS/TLS) depends on the deployment environment and must be verified there.

2.5 Constraints. The current repository uses FastAPI and SQLite. File/image storage and backup behaviour must be documented for the deployed configuration. POS billing and supplier-side external procurement integration remain outside this

release.

2.6 Assumptions/dependencies. Users have valid accounts and role assignments; suppliers/categories/branches are configured before dependent workflows; purchase-order receiving refers to valid components and suppliers; database transactions are available for atomic stock changes.

3. External interface requirements

3.1 User interface. The browser UI shall provide navigation for inventory, suppliers, branches, purchase orders, stock movements, reports and dashboard functions as applicable to the user role. Forms shall show validation feedback. Inventory filters shall support available category/supplier/stock criteria. Theme preference shall persist across reloads where the user selects a theme.

3.2 Hardware interfaces. Standard keyboard, mouse and touchscreen/tablet input. Barcode scanner and printer are not required for the current release.

3.3 Software interfaces. The frontend communicates with the FastAPI REST API. The API uses SQLAlchemy persistence. Documented endpoints include /api/auth/login, /api/inventory, /api/branches, /api/suppliers, /api/categories, /api/purchase-orders, /api/inventory/{id}/movements, /api/reports/availability and /api/reports/system-availability.

3.4 Communications. API requests and responses use HTTP semantics and JSON for application data. HTTPS/TLS must be enabled in production deployment; local development may use localhost and must not be represented as proof of production TLS.

4. Functional requirements — original baseline

The original functional requirement IDs are retained. The summary below preserves their intent; detailed original acceptance criteria remain applicable.

ID	Requirement (shall)	Acceptance summary
F-001	Display active inventory with name, SKU, category, quantity, unit price, supplier and reorder level.	Inventory list displays correct active records.
F-002	Search components by name, SKU or category.	Matching results returned; original target is within 2 seconds.
F-003	Filter components by category, supplier and stock status.	Results match selected filter criteria.
F-004	Sort components by quantity, price or date added.	Ordering matches selected sort key and direction.
F-005	Allow authorized users to update component details.	Updated values persist and appear in inventory.
F-006	Add components with required fields.	Valid component is saved and listed.
F-007	Generate a unique component ID/SKU.	New SKU is unique and not reused.
F-008	Validate required fields and invalid values.	Invalid submission is rejected with clear feedback.
F-009	Prevent duplicate SKU entries.	Duplicate SKU is rejected without corrupting existing data.
F-010	Allow optional component image upload.	Accepted image is stored and associated with component.
F-011	Allow authorized component deletion/discontinuation.	Component is removed from active inventory as specified.
F-012	Require confirmation before deletion.	User can confirm or cancel the operation.
F-013	Restrict deletion to authorized roles.	Unauthorized staff request is rejected.
F-014	Record deletion audit information.	Audit record contains user, timestamp and component.
F-015	Generate availability classification report.	Components classified as in-stock, low-stock or out-of-stock.
F-016	Filter availability report.	Report reflects selected filters.

ID	Requirement (shall)	Acceptance summary
F-017	Export availability reports.	Supported export formats are generated from report data.
F-018	Highlight low-stock components.	Low-stock items are visibly identified.
F-019	Display dashboard inventory summary.	Summary totals reflect persisted inventory.
F-020	Display stock value/summary information.	Displayed calculations match underlying records.
F-021	Display dashboard charts/recent inventory information.	Charts and recent items reflect current data.
F-022	Display low-stock dashboard alerts.	Low-stock alerts reflect configured reorder thresholds.

5. Functional requirements — added and expanded in version 1.1

The following IDs extend the original baseline. Where a feature is present in the repository but a detailed business rule was not explicit in the source SRS, the rule is stated as a requirement for team confirmation before final approval.

ID	Feature	Requirement and acceptance criteria
F-023	Supplier catalog management	The system shall maintain supplier records and expose authorized supplier creation and retrieval. Acceptance: valid supplier persists; invalid data is rejected; catalog is available to component workflows.
F-024	Branch management	The system shall provide branch records and branch management operations. Acceptance: authorized user can create/list/update branch data; invalid or unauthorized requests are rejected.
F-025	Component image lifecycle	The system shall support image preview, replacement and removal in addition to original upload. Acceptance: supported image formats within the configured 5 MB limit are accepted; unsupported/oversized files are rejected; removal/replacement is reflected in the component view.
F-026	Add-component form usability	The system shall provide clear required-field indicators, inline validation, responsive layout and cancellation/reset of unsaved changes. Acceptance: invalid fields are identified; cancel resets unsaved form values.
F-027	Dynamic supplier/category filtering	The system shall populate supplier and category filter choices from persisted data. Acceptance: newly added supplier/category values become selectable and combined filters return matching records.
F-028	Stock movement history	The system shall record stock-in and stock-out movements with component, movement type, quantity, reason, timestamp and responsible user. Acceptance: valid stock changes persist with history; insufficient stock and invalid quantities are rejected; authorized roles are enforced; failed operations roll back.
F-029	Purchase order management	The system shall allow authorized users to create purchase orders for existing suppliers, add multiple components and quantities, view/update order status, receive orders partially or fully, and retain receipt history. Acceptance: receiving updates inventory once only; duplicate submissions are idempotently handled; failures roll back.
F-030	Availability report error handling and system reporting	The system shall provide availability reporting and handle missing monitoring/report data without exposing internal errors. Acceptance: missing records and unexpected failures return controlled responses; scheduled maintenance and monitoring coverage are treated consistently.
F-031	Persistent theme preference	The system shall allow users to switch between light and dark presentation modes and retain the selected preference across reloads. Acceptance: selected theme is applied consistently across supported UI surfaces.
F-032	Consistent API error responses	The system shall return structured, appropriate error responses for validation, missing resources, authorization failures and unexpected errors. Acceptance: clients receive stable status codes and readable messages without sensitive internals.

6. Non-functional requirements

The original measurable non-functional requirements remain in force. Their measurement results must be recorded in the Test Summary Report; do not mark them passed solely because a test exists.

ID	Quality attribute	Measurable requirement
NF-001	Performance	Dashboard and inventory list pages load within 3 seconds for at least 90% of requests under normal load.
NF-002	Availability	Monthly system availability is calculated from monitoring records, with scheduled maintenance and coverage limitations handled explicitly.
NF-003	Usability	A new staff user can add or search for a component within five minutes without formal training.
NF-004	Scalability	System supports a test dataset of 10,000 components and 20 concurrent users without unacceptable degradation; record actual environment and measurements.
NF-005	Maintainability	UI, API/application logic and data access remain separated, documented and testable.
NF-006	Portability/responsiveness	Web UI remains usable at desktop viewport width of at least 1280 px and tablet viewport width of at least 768 px.

7. Security requirements

The original security requirements are retained. Their controls apply to the new branch, supplier, movement, purchase-order and image-management endpoints as well as the original inventory endpoints.

ID	Control	Requirement
SR-001	Authentication	Protected pages and APIs require an authenticated user; unauthenticated API access is rejected.
SR-002	Password storage	Passwords are stored as salted one-way hashes; plaintext passwords are not stored.
SR-003	Role-based authorization	Admin/Manager and Staff permissions are enforced server-side; restricted operations return an authorization failure.
SR-004	Transport security	Production client-server communication uses HTTPS with TLS 1.2 or later.
SR-005	Session handling	Inactive sessions terminate after the specified 15-minute inactivity period.
SR-006	Input and output safety	Validate/sanitize untrusted inputs and safely encode output to mitigate injection and XSS.

8. Data requirements and business rules

Inventory quantities shall not become negative through stock-out or purchase-order receiving operations. Stock changes and their movement records shall be committed consistently. Purchase-order receipt operations shall prevent duplicate stock increments. Supplier and branch records referenced by dependent entities must exist. Image upload validation shall enforce the configured file-size and format restrictions. Audit/history records shall identify the acting user and event time where specified.

The database design and constraints shall preserve referential integrity for components, suppliers, branches, purchase orders, purchase-order items, receipts and stock movements. Exact deletion/retention rules for suppliers, branches and historical records require team confirmation where not specified by existing implementation.

9. Acceptance, verification and traceability

Each requirement shall be verified through one or more of: automated unit/integration/security tests, API-level checks, UI smoke tests, performance measurements or document/code review. The companion STP provides detailed test cases and RTM. A requirement is accepted only when its acceptance criteria are met and evidence is recorded.

The original requirements F-001 to F-022, NF-001 to NF-006 and SR-001 to SR-006 are preserved. Version 1.1 extends the functional set through F-032. New requirement IDs must be reconciled with the GitHub backlog before final sign-off; IDs F-023 to F-032 are the proposed consolidated numbering for this revision.

10. References

Original EIMS SRS v1.0 dated 06-09-2026; EIMS Software Test Plan v1.0 dated 16-09-2026; Team 9 SAD v1.0 dated 16-09-2026; repository README, docs/API.md and docs/ARCHITECTURE.md; repository issues and merged pull requests relating to supplier/branch management, image management, stock movement history, purchase orders, availability reporting, theme/UI improvements and API error handling.